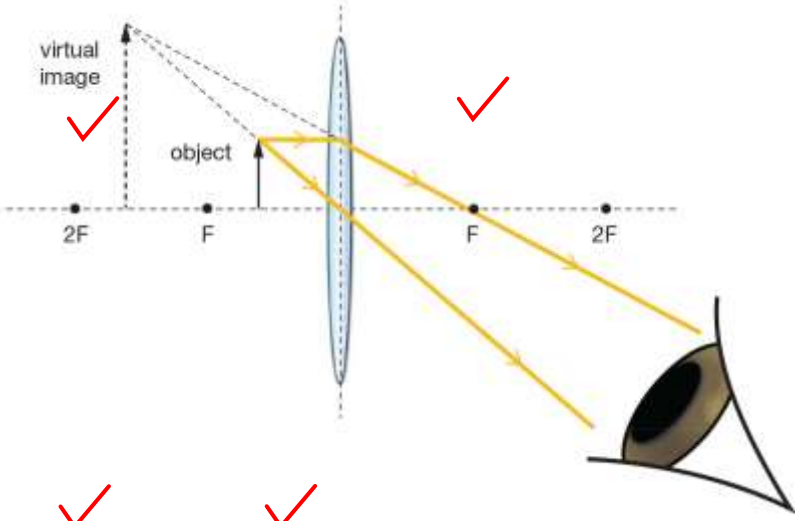
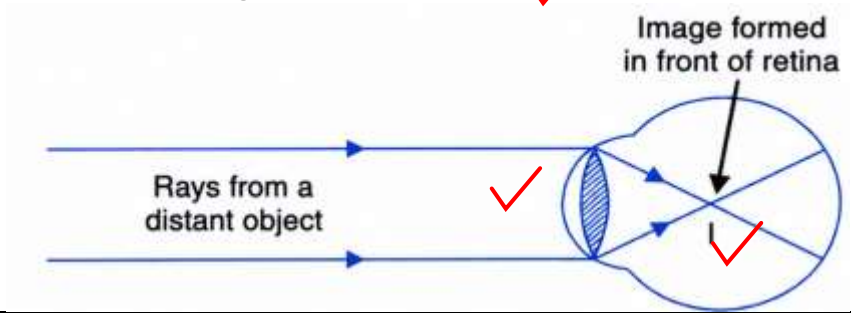
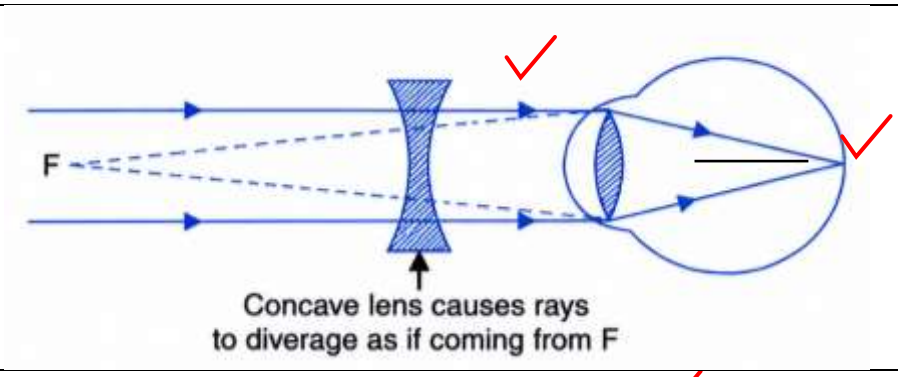
SCORING GUIDE
MATIGO MOCK EXAMINATIONS BOARD
UCE PHYSICS THEORY 2026

Item 1	(a) An object is placed between the optical centre of the lens and its principal focus.  An upright, magnified and virtual image is formed.	
(b)	The people are short sighted. It is caused by the eye ball being too long leading to the formation of an image before the retina.  	
	It is corrected by wearing eye glasses with concave lens.	
	 	
(c)	$v = f\Lambda$ $3 \times 10^{8} = f(3 \times 10^{-7})$ $f = 1 \times 10^{15} Hz$ The rays from the sun are harmful because $1 \times 10^{15}\text{Hz} > 7.5\text{x}1014\text{Hz}$	
(d)	• Laser light is a highly focussed, intense beam of coherent light. • Energy is supplied to atoms in an active medium through a process called pumping.	

	• Electrons absorb energy and more to higher energy levels. • Some excited electrons remain temporary in a meta state. • Incoming protons trigger stimulated emission, producing identical protons. • The protons are amplified between mirrors and emerge as narrow intense laser beam.	
	• Lasers are used in operations to correct eye defects. • Lasers cut tissues with minimal bleeding as they seal blood vessels while cutting. • They are used to destroy cancer cells. • They are used in optical fibres to view internal organs. • They are used in dermatology to remove tattoos.	
Item 2	• The teacher meant that stars have a life cycle. • A star starts out as a large cloud of dust and gas called nebulae. • Ave to gravitational collapse, nebula collapses further leading to an increase in pressure leading to the formation of a protostar. • As pressure increases, so does temperature. When temperature reaches 10 million Kelvin, nuclei fusion starts leading to the formation of a main sequence star. • After fuel is exhausted, small stars become red giants and white dwarfs while large ones become red super giants and explode as supernovas. • The remnants of massive starts form either neutron stars or black holes.	
	Stars appeared different from each other in terms of brightness and colour. The difference in brightness is due to; • Size of the stars. Large stars are brighter than small ones. • Distance of stars from the earth. The nearer the star to the earth, the brighter it is. The difference in colour is due to; • The temperature of the star; Hotter stars appear white and blue while cooler ones appear red and yellow. • Amount of hydrogen in the star. Young stars have more hydrogen and are therefore hotter than older ones.	
(b)	• The program was aired using communication satellites. • Images and audios are captured using cameras and microscope. • They are changed into a format suitable for transmission. • They are sent to a ground station. • They are uplinked to a satellite in the geostationary orbit. • The waves are amplified and sent back to the earth to be received by a satellite dish facing the satellite outside the earth.	
(c)	The earth moves in two ways; • The earth spins on its axis. As a result of this, day and night occur. • The earth also revolves around the sun leading to the formation of seasons.	
Item 3	• Electricity is generated as a result of nuclear fission when uranium – 235 is bombarded with neutrons in a nuclear reactor according to the equation; • $^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{142}_{b6}Ba + {}^{92}_{36}Kr + 3{}^{1}_{0}n + Energy$	

- The energy produced is in form of heat energy and it is used to heat water.
- When the water boils, it produces steam at high pressures.
- The steam is used to run turbines connected to generators that convert mechanical energy to electric energy.

To ensure safety.

- All workers should wear safety wear like lead jackets.
- All workers should have radiation monitors.
- All radioactive materials should be kept in lead containers.
- Radioactive waste should be kept when its activity is below the background radiation.

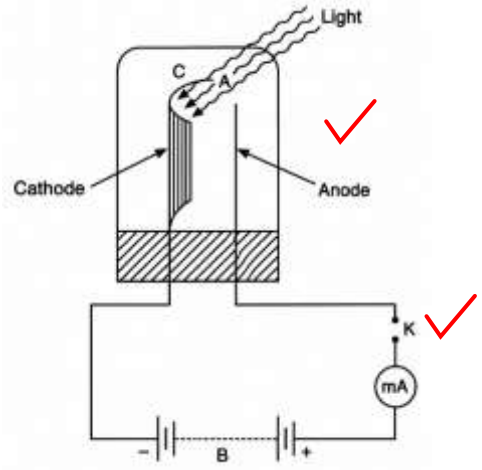
(b) Materials with long half-lives are used because they decay slowly, producing energy over a long time at a steady rate reducing the chance of nuclear accidents.

(c)

Activity	Time taken
960	0
480	6
240	12
120	18

The material should be used for only 18 hours.

(d) The automatic doors work as a result of photoelectric emission.

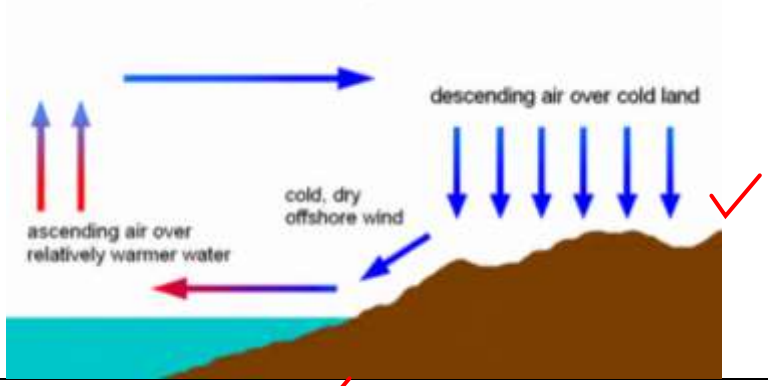


- When ultraviolet light mode incident on a cathode it produces electrons by photoelectric emission.
- The electrons move from the cathode to the anode closing the circuit.
- As a person approaches the door, they cut off the ultra – violet light.
- The photo cell has an automatic switch that opens the door whenever electrons are not produced.

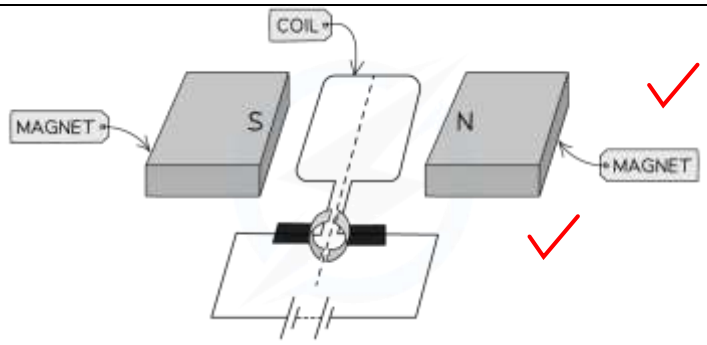
Item 4 (a) From the principle of moments, the sum of the clockwise moments is equal to the sum of the anti-clock wise moments.

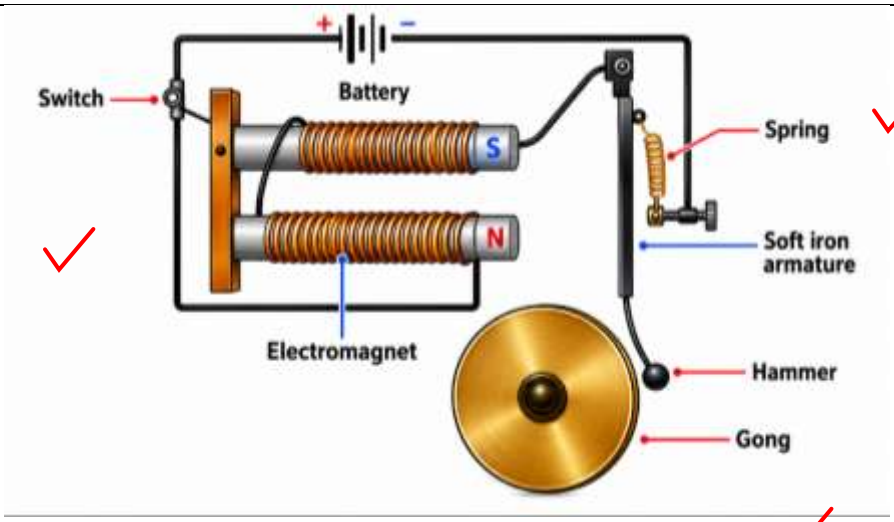
$$M_1 \times d_1 = m_2 \times d_2.$$

$$M_1 \times 20 = 10 \times 2000$$

	$M_1 = \frac{10\ x\ 2000}{20}$ $= 1000g = 1kg$ The fish did not meet the clients' requirement.	
(b)	At the base of a mountain, atmospheric pressure is equal to the blood pressure. As the people move up the mountain, the atmospheric pressure reduces. This creates a pressure difference between the person and the environment leading to the breaking of blood vessels in the nostrils of the people.	
	A wooden container is a poor absorber of heat ensuring heat doesn't reach the fish. White is also a good reflector of heat reducing heat gain in the container. The ice works as a preservative for the fish since it helps reduce bacteria build up in the fish.	
	At night the temperature of the environment is low. This leads to the water being warmer than the land since the land cools faster than the water. The air above the water rises, and is replaced by air from the land ie land breeze. As the process continues, convectional cements are formed.  	
	• Land breeze leads to the formation of rain fall used for domestic use. • The breeze provides wind for cooling coastal towns. • The breeze dries clothes that are worn by people.	
Item 5 (a)	The chimney allows warm stale air to exit from the building since it becomes less dense after being heated. • The cooking stoves should be lined with clay to reduce the amount of fuel used when cooking. This is because clay is a poor conductor of heat reducing the rate of heat loss. • A lift pump operates solely on atmospheric pressure which can only support a water column of maximum depth 10m. A force is therefore preferred to the lift pump since it can draw water from bigger distances.	
(b)	The water was changed to ice as a result of freezing. • When water is put into the refrigerator, the volatile liquid in the copper coils draws heat from the water. • The volatile liquid changes to gas drawing latent heat of vaporization from liquid. • The gas goes to the compressor where it is turned back to liquid.	

	• As the process continues, more heat is drawn from the water making the water change to ice.	
(c)	Assuming no heat loss to the environment, Heat lost by hot water = Heat gained by ice + Heat gained by saucepan. $M_hC_w\ (100-\theta) = m_i c_i(0--20) + m_i l_f + +m_i c_w\theta + m_p c_p(\theta--20)$ 1litre of water has a mass of 1kg. $(1)4200(100-\theta)=(0.8\times2100\times20)+(0.8\times340{,}000)+(0.8\times4200\theta)+200(\theta+20)$ $4200(100-\theta) = 33600 + 272000 + 3360\theta + 200\theta + 4000$ $420{,}000 - 4200\theta = 309600 + 3560\theta$ $420{,}000 - 309600 = 3560\theta + 4200\theta$ $110400 = 7760\theta$ $\theta = 14.23^0C.$ The water did not attain the desired temperature.	
Item 6 (a)	$Kettle\ \frac{2000}{1000}\times 1 = 2Kwh$ $Reteragram\ \frac{300}{1000}\times 24 = 7.2KWh.$ $Water\ heater\ \frac{2500}{1000}\times 2 = 5KWh$ $Bulbs.\frac{10}{1000}\times 8\ \times 3 = 0.24Kwh.$ $Television\ \frac{150}{1000}\times 5 = 0.75Kwh.$ Total number of Kwh used; $= 2 + 7.2 + 5 + 0.24 + 0.75 = 15.19Kwh.$ Cost = Number of Units x Cost per unit. $= 15.19\ x\ 1000.$ $= 15{,}190/=$	
(b)	The socket melted due to overloading of the socket or poor electrical contact. When a large current is drawn from the socket, electrical energy is converted into heat energy leading to the production of too much heat. Additionally, if the plug connections are loose, spanks are produced due to the absence of smooth flow of current leading to the production of heat which melts the socket.	
(c)	The fan works on the principle that a current carrying conductor placed in a magnetic field experiences a force.	

	 A DC motor has a rectangular coil made of insulated copper wire, permanent magnets and split ring. When current flows the coil, the coil becomes magnetized. The magnetic field around the coil interacts with the magnetic field of a permanent magnet. A couple is developed on the coil leading to the coil rotating. When the coil reaches its vertical position, the coil loses contact with communicators but it continues moving due to inertia. The current will be reversed but the coil will continue moving in the same direction. The fan can be safeguarded by; • Installing a fuse to break the circuit in case of excess current. • Using a circuit breaker to automatically disconnect power during overload. • Carrying out regular maintenance to reduce friction brought about by rubbing of the moving parts.	
Item 7	Total internal resistance for the cells = 4 x 0.5=2Ω. Total emf = 4 x 1.5 = 6.0V $From\ P = \frac{V^2}{R}$ $$Lamp\ resistance\ R = \frac{V^2}{P} = \frac{6^2}{3} = 12\Omega.$$ Total resistance in circuit = 12 + 2 = 14Ω. From V = 1R $$6 = \mathrm{I} \times 14$$ $$\mathrm{I} = 0.43\mathrm{A}.$$ Voltage across the lamp. $$\mathrm{V} = 1\mathrm{R}$$ $$\mathrm{V} = 0.43 \times 12 = 5.14\mathrm{V}.$$ The light is dim because the lamp receives less voltage than it is meant to operate on. The amount of light can be increased by using cells of lower internal resistance.	
(b)	Namara should install an electrical bell.	

	 An electric bell has a switch, electromagnet hammer gong etc. When the circuit is closed, current flows through the circuit. This current induces a magnetic field around the electromagnet. The magnetised electromagnet attracts the iron producing sound. It breaks the electrical contact making the circuit to be open. The electromagnet ceases to be one enabling the spring to return its iron armature to its position when the circuit closes again. For as long as the switch is on, magnetisation and Demagnetisation will continue happening, so the bell will keep ringing.	
(c)	Earthing ensures none of the users is shocked by electricity in case the live wire touches any of the metallic parts of the bell. It does this by providing an alternative path for current to the ground.	

END